

Article

Health Behaviors Among Students and Their Association with Stress, Student Burnout and Study Engagement

Nils Olson *, Renate Oberhoffer , Barbara Reiner  and Thorsten Schulz

Chair of Preventive Pediatrics, TUM School of Medicine and Health, Technical University of Munich, Am Olympiacampus 11, 80809 Munich, Germany

* Correspondence: nils.olson@tum.de

Abstract: The prevalence of stress and burnout in higher education exceeds rates of the general working population, affecting students' health and increasing university drop-out rates. Study engagement—a positive state of energy, dedication, and absorption—acts as a protective factor. A healthy lifestyle is neglected by many students, especially early in their studies. Since habits like diet, physical activity, weight management, and substance use impact mental health and resilience, their role in stress and burnout should be explored. In a cross-sectional online survey of 1955 German university students, sociodemographic data were collected and perceived stress, study engagement, and burnout were assessed. Associations with health habits, including diet, fitness, and tobacco and alcohol use, were examined. Physical activity was measured via the International Physical Activity Questionnaire, and alcohol consumption was measured via the Alcohol Use Disorders Identification Test. Among the students, 73.4% experienced moderate to high stress, with females reporting higher levels. Almost one-third had frequent burnout symptoms, while less than half (44.5%) reported high study engagement, with no sex differences. Active lifestyles, healthy diets, and fitness were linked to stress, burnout, and engagement. Perceived fitness and diet had the strongest associations with stress ($d = 0.198$ – 0.906), burnout ($d = 0.277$ – 0.483), and engagement ($d = 0.218$ – 0.272), while BMI ($d = 0.089$ – 0.186), sitting hours ($d = 0.172$ – 0.203), and physical activity ($d = 0.096$ – 0.141) played a comparatively minor role, regarding their small effect size. Personal habits are significantly linked to burnout and engagement, possibly by enhancing resilience. Health literacy regarding study demands is key for effective prevention and targeted student health management. More objective fitness data could further validate these findings.

Keywords: stress; student burnout; study engagement; health habits; lifestyle; university; students



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1. Introduction

Chronic distress and burnout have long been subjects of research due to their profound impact on occupational functioning and overall well-being within workplace settings [1,2]. These conditions not only undermine employees' productivity, health, and quality of life, but also have broader implications for economic performance and public health systems [3].

More recently, stress and burnout have been identified as a major problem amongst not only the working population but also students. The term 'student burnout' has been developed for this specific target group at high schools and universities alike. Stress and burnout among students are strongly linked to ill health, reduced well-being and self-esteem, and a higher prevalence of university drop-out rates and even suicidal ideation [4–6].

Student burnout is characterized by a state of diminished capacity to engage, accompanied by emotional and physical exhaustion, along with depersonalization, a sense of decreased coping ability, and cognitive slowdown [7]. It is marked by a combination of emotional exhaustion (EE), cynicism (CY), and a perceived reduction in academic efficacy (RAE).

From occupational research, a contrastive factor for burnout has been identified: work engagement. It has been found to be applicable for educational systems like schools and universities [8–14]. ‘Study engagement’ or ‘academic engagement’ represents the positive counterpart to student burnout. In fact, student burnout is often seen as the result of gradual erosion of academic engagement [12]. Study engagement is defined across three dimensions: vigor (i.e., having high energy levels while studying), dedication (i.e., viewing one’s studies as important and meaningful), and absorption (i.e., being fully immersed in one’s studies). Together, these dimensions foster a positive, fulfilling mental state, sustained energy, and positive study-related emotions. High levels of engagement are associated with positive health outcomes [15].

The student population has been found to be prone to significant stress and frequent burnout symptoms as well as low prevalences of functional study engagement [16].

The specific stage of life is attributed to various changes in personal and career-oriented areas of life on top of the specific demands that studying itself causes [17–19]. These changes entail the transition from late adolescence to young adulthood, major shifts within the social environment and social roles, the detachment from family and parental home, the pursuit of educational and occupational choices, changes in romantic status, and often a double burden of academic and occupational liabilities [20]. The high prevalence of mental health disorders, stress, and suicidal ideation among students [4,21] highlights the intense pressures of major life transitions, compounded by the demands of university and work life.

Being a more recent topic, studies on student burnout and study engagement are scarce, but it can be concluded that the prevalence of burnout is high in the student population, with rates ranging from 12% to more than 70% [16,22–26], thus exceeding the rates of workers in the medical fields [27]. Respectively, with only a few studies pointing to the prevalence of study engagement, it can only be assumed that barely half of the students are highly engaged within their study courses [16,28–30].

Next to sociodemographic factors that have produced mixed results in regard to the role of gender and age [16,23,26,31,32], some studies have been performed on additional contributing factors, often relying on the framework of the study–demands–resources model (SDR) [33]. Those studies are limited and have been performed with a variety of different methods. However, some conclusions can be drawn from them, namely associations between burnout and engagement. Overall, students with greater psychological well-being seem to exhibit greater vigor, dedication, and absorption. Self-efficacy, defined as the belief in one’s capabilities to execute tasks successfully, is also a strong predictor of academic engagement [34]. Further factors involve structural conditions, such as study satisfaction, supervisor support, scope of action, semester progression and academic workload, as well as faculty affiliation [16], but also person-oriented parameters, like mental health status (e.g., anxiety), marital status, and psychological flexibility [16,30,32,35–37].

Health habits, on the other hand, have not been systematically included in this model yet, despite the fact that health behavior plays a vital role in mental and physical health [38,39]. Health habits refer to actions that are regularly performed over a long period of time, such as, but not limited to, regular and sufficient physical activity levels, a healthy diet, and avoidance of excessive sedentary time, tobacco usage and unhealthy amounts of alcohol. Interestingly, a decline in healthy lifestyle habits has been observed

among many college students, especially upon entering university [40]. It has been shown that the transition to university is characterized by increased autonomy for students due to the detachment from the parental home. Existing health competences are often not sufficiently established to prevent the development of unhealthy habits [41]. Accordingly, physical activity (PA) has been shown to decrease amongst students, particularly amongst those within their first year of studying [41–45]. In a German-wide investigation, less than half of the students regularly achieve even the lower end of the physical activity recommendation (according to the World Health Organization) of 150 min per week, and sedentary time increases [46]. Other negative behaviors observed in university students include the increased consumption of alcohol and tobacco [47]. On top of that, university students are prone to engaging in undesirable dietary habits, often resulting in weight gain [48,49]. These lifestyle factors, although linked to the onset and chronification of mental disorders [50], are not disease-specific but instead contribute to the risk of developing physical and mental health problems and contribute to the impairment of social, occupational, and academic performance [24,51,52].

A clear link between PA, diet, unhealthy alcohol and tobacco consumption, and obesity to mainly (mental) ill health is well established [53–57]. The link to study demands, stress, and also good health in the general population, however, is not as well understood, because most studies focus on pathologies, or secondary or tertiary prevention rather than primary prevention among healthy (young) adults.

In regard to stress and burnout, there are not many studies focusing on students, and, to date, there have not been any studies carried out on the different intensities of PA (low, moderate and vigorous) on students of different subject fields [58].

The goal of this study is therefore as follows:

- (a) To quantify health behavior in a large sample of students of different academic study fields at a technical university in Germany;
- (b) To detect interrelations to stress;
- (c) To identify associations to study engagement and student burnout and therefore adding to the study–demands–resources model.

2. Materials and Methods

This study was designed as a cross-sectional survey, involving an online questionnaire that addressed 18 different health dimensions. Depending on the responses to filter questions, the survey included up to 89 items. The majority of the survey blocks were composed of internationally standardized and validated instruments. Additional data on anthropometric, sociodemographic, and study-related aspects were also gathered. For the purposes of this research, data on age, sex, body mass index, smoking behavior, alcohol consumption (AUDIT-C), perceived fitness level and healthiness of the diet, the Utrecht Work Engagement Scale for Students (UWES-S 9), the Maslach Burnout Inventory Short Form for Students (MBI-SS), and the Perceived Stress Scale (PSS-10) were analyzed.

From November 2019 to January 2020, students at the Technical University of Munich ($N = 45,876$) were invited via email to participate in the online survey. Only those aged 18 or older were eligible. Participants were fully informed about the study's aims and provided written consent before participating. The survey yielded responses from 4720 students, resulting in a response rate of 10.3%, with 54.8% being female ($n = 2588$), 44.7% male ($n = 2108$), and 0.5% nonbinary ($n = 24$).

2.1. Student Burnout—Maslach Burnout Inventory (MBI-SS)

Burnout symptoms were measured using the MBI-SS, which was designed for academic settings [12]. The inventory assesses three dimensions: emotional exhaustion (EE),

which reflects fatigue due to study demands and represents the individual stress component; cynicism (CY), indicating a mental distancing from studies and detached responses to peers and teachers, representing the interpersonal component; and reduced academic efficacy (RAE), indicating a sense of decreased competence and productivity and a diminished sense of accomplishment, representing the self-evaluation component [9,59,60]. Each dimension comprises three items, rated on a 7-point Likert scale from 0 (never) to 6 (always). Higher scores signify more frequent symptoms. Each dimension was analyzed independently, using the mean score to assess symptoms. Participants reporting symptoms ‘frequently’ (at least once per week) were grouped accordingly. The MBI-SS has been internationally validated for measuring student burnout [61,62]. In our study, Cronbach’s Alpha values were 0.831 for EE, 0.867 for CY, and 0.746 for RAE.

2.2. Study Engagement—Utrecht Work Engagement Scale Student Version (UWES-S 9)

The UWES-S 9 [12] evaluates engagement levels in students through three dimensions: vigor (high energy levels while studying), dedication (finding one’s studies important and meaningful), and absorption (deep immersion in one’s studies). Items are scored similarly to those in the MBI-SS. This scale has international validation [11,12,36,63]. Engagement was calculated as the mean score of all three dimensions, ranging from 0 to 6, with higher scores indicating greater engagement. Students with scores above 3.5 were classified as ‘highly engaged’.

The MBI and UWES-S have both been proven to maintain their validity, reliability, and consistency in their respective German versions, which have been used in our study [61,64].

2.3. Perceived Stress Scale—PSS-10

Stress assessment in college students was conducted using the Perceived Stress Scale (PSS-10), which gauges perceived stress and reactions to stressful situations. It correlates with numerous psychological and physiological measures. The PSS-10, selected for its validity and reliability [65], which has been confirmed for its German version [66], includes 10 items rated on a 5-point Likert scale, focusing on the last four weeks’ stressful experiences and the respective reactions. Negative responses are reverse scored to ensure correct interpretation, with higher overall scores indicating higher stress levels. Scores range from 0 to 40 [67]. Published reference scores allow for comparison across different age groups and sexes [68]. Stress levels were categorized as low (0–12), moderate (13–26), and high (27+) according to arbitrary thresholds that have been used in previous studies before [69,70]. This sample’s internal reliability (Cronbach’s α) was 0.890.

2.4. Health Behavior

2.4.1. Physical Activity and Sedentary Behavior—IPAQ

The International Physical Activity Questionnaire (IPAQ) was used to measure levels of moderate and vigorous PA, walking time, and sedentary behavior. The IPAQ is a commonly used standardized self-report questionnaire and is well validated in adult populations [71]. For the IPAQ, participants were asked to count the days of moderate and vigorous PA per week and state the number of minutes of one of those sessions. The same was assessed for “walking” and minutes spent sitting. The analysis was performed following the instructions in the official manual [72]. After calculating the number of minutes of each activity per week, each category is multiplied by the number of metabolic equivalents (METs) (3.3 for walking, 4.0 for moderate activities, and 8.0 for vigorous activities). The weekly METs are then summed up to provide a total sum score and a sum score for each category. In addition, minutes spent sitting were also assessed. The classification of PA was determined in three different categories, namely ‘low’, ‘moderate’, and ‘high’, according to the criteria in the manual [72].

2.4.2. Perceived Fitness, Perceived Healthiness of Diet, Alcohol and Tobacco Consumption

Participants were also asked how they estimate their own fitness level and how healthy they think their diet is. Both could be checked on a 10-point Likert scale, with higher values indicating a higher fitness level or healthier diet, respectively. Smoking behavior was categorically assessed ('yes'/'no'/occasional). Alcohol consumption behavior was assessed using the 3-item Alcohol Use Disorders Identification Test (AUDIT-C), a reliable and valid instrument to identify alcohol misuse [73]. The resulting sum score ranges from 0 to 12, with higher numbers being associated with an increased likelihood of hazardous alcohol usage affecting the individual's health negatively.

2.5. Statistical Analysis

Descriptive data are reported as mean (M) $\pm$ standard deviation (SD) for metric variables and as frequencies and percentages for categorical variables. Pearson's correlation analysis examined relationships among metric variables.

The high internal correlation among UWES-S 9 dimensions (Vigor, Dedication, Absorption (0.74–0.87) led to using a one-factor structure for this scale [74], while maintaining the three-factor structure for the MBI-SS due to the distinctiveness of the subscales. Differences between two groups were analyzed using Student's *t*-test, with Cohens *d* (*d*) indicating the effect size. To examine distribution differences between categorical variables contingency analyses were conducted. The Chi-square test was used for symmetric tables, and Cramer's *V* served as an asymmetric measure of associations. ANOVA was used for comparing more than two groups, and Welch-ANOVA was employed when homogeneity was violated. GT2 Hochberg's testing was applied for pairwise post hoc comparisons, because of its applicability for different group sizes. Due to the small number of non-binary participants, they were excluded from sex-differentiating calculations but not from descriptive statistics.

Statistical significance was set at $p \leq 0.05$. All analyses were performed using SPSS 29.0 (IBM Inc., Armonk, NY, USA).

3. Results

Graduate and undergraduate students who completed all relevant questionnaires were included, leading to a total sample of $N = 1955$, of which 40.1% ($n = 783$) were male, 59.6% were female (1165), and 0.4% were non-binary ($n = 7$). Ages ranged from 18 to 55 years with an average of 22.4 ± 3.22 years. The semester progress ranged from the first semester to the 25th semester, with the majority of the students being in their first to third semester (59.3%). Most of the students (55.4%) were in their bachelor's degree program, 29.9% in their master's programs, 14.4% were in a state examination program, and 0.3% were doing their diploma.

Participants scored 18.3 ± 6.90 on the PSS-10. Thus, 26.5% of the students were classified as having "low levels" of stress, 61.3% as "moderate", and 12.1% as having "high levels" of stress. Females had significant higher scores on the PSS (19.3 ± 6.68) in comparison to male students (16.7 ± 6.87) ($t(1946) = -8.323$; $p < 0.001$, $d = 0.385$). Compared to an 18-to-29-year-old reference population (14.2 ± 6.2) [68], a significantly greater score on the PSS was found among all students of this sample ($t(1954) = 26.108$; $p < 0.001$).

The students scored 2.6 ± 1.53 on the burnout dimension of EE, 1.1 ± 1.44 in CY, and 1.9 ± 1.54 in RAE. While no difference has been observed in EE and CY between male and female students, women scored significantly higher in the burnout dimension of RAE (2.0 ± 1.55) than men (1.80 ± 1.50) ($t(1946) = -2.879$; $p = 0.004$, $d = 0.133$).

A total of 28.1% ($n = 549$) of the students had frequent symptoms of burnout in at least one of the three dimensions, while 22.7% ($n = 443$) suffered from frequent symptoms of EE,

7.2% ($n = 141$) from frequent CY and 13.6% ($n = 265$) frequent RAE. In total, 3.9% ($n = 76$) had frequent symptoms in all three dimensions simultaneously.

Overall, 44.5% ($n = 869$) of the students were highly engaged, with no differences between male and female. 5.7% ($n = 112$) of the sample was highly engaged while simultaneously having frequent symptoms of burnout in at least one dimension.

BMI classification revealed 81.7% of the students being within the norm-BMI-range of 18.5–25.0 kg/m². 6.4% were underweight, 10.1% overweight and 1.8% fell in one of the three categories of obesity. Men (22.9 ± 3.01 kg/m²) had a significant higher BMI than women (21.4 ± 2.97 kg/m²) ($t(1939) = 10,102$; $p < 0.001$, $d = 0.468$).

While 1.9% of all men were underweight, 9.3% of women had an BMI less than 18.5 kg/m². On the other hand, 13.9% of men were overweight, while only 7.4% of women were. Overall, 2.7% of men and 1.2% of women were classified as obese (Table 1).

Table 1. BMI categorization according to gender.

BMI Classification	Underweight	Normal Weight	Overweight	Obesity I	Obesity II	Obesity III	Total
Male	1.9% $n = 15$	81.5% $n = 634$	13.9% $n = 108$	2.3% $n = 18$	0.3% $n = 2$	0.1% $n = 1$	100.0% $n = 778$
Female	9.3% $n = 108$	82.1% $n = 955$	7.4% $n = 86$	0.9% $n = 11$	0.0% $n = 0$	0.3% $n = 3$	100.0% $n = 1163$
Non-binary	14.3% $n = 1$	57.1% $n = 4$	28.6% $n = 2$	0.0% $n = 0$	0.0% $n = 0$	0.0% $n = 0$	100.0% $n = 7$
Total	$n = 124$	$n = 1593$	$n = 196$	$n = 29$	$n = 2$	$n = 4$	$N = 1948$

The correlation analysis showed strong associations between all three burnout dimensions and a moderate to strong relationship between stress and burnout. Study engagement was moderately negatively correlated to stress, EE and RAE, and CY strongly negatively related to study engagement.

Subjective fitness and perceived healthiness of the individuals' diet are weakly negatively correlated to stress as well as to each burnout dimension, while being weakly positively associated with study engagement. Daily hours of sitting and EE are weakly correlated (Table 2).

Women sat an average of 6.0 ± 4.03 h per day and thereby significantly less than men, whose sitting time was 6.8 ± 3.44 h a day ($t(1920) = 4.157$; $p < 0.001$; $d = 0.193$). While men were engaged in more weekly vigorous activity of 1622.2 ± 1412.55 MET compared to women's average of 1456.6 ± 1402.51 METS ($t(1946) = 2.548$; $p = 0.005$; $d = 0.118$), women were walking significantly more (650.3 ± 762.65 METs vs. 588.3 ± 621.57 METs) ($t(1873.050) = -1.966$; $p = 0.025$; $d = 0.087$).

Overall, 42.5% of the students had a "normal" drinking behavior according to the AUDIT-C. A total of 21.4% were at increased risk for hazardous drinking and 36.1% displayed drinking behavior that could be classified as an alcohol use disorder with no difference between males and females according to Cramer's V analysis.

In total, 4.1% of the students were smokers while another 10.0% described themselves as occasional smokers. According to Cramer's V analysis, the distribution between male and female differed significantly ($p = 0.048$) (Table 3).

Table 2. Correlation analysis.

	Variable	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1.	Perceived Stress	18.3	6.90														
2.	Emotional Exhaustion	2.6	1.53	0.598 **													
3.	Cynicism	1.1	1.44	0.454 **	0.550 **												
4.	Reduced Academic Efficacy	1.9	1.54	0.581 **	0.581 **	0.569 **											
5.	Study Engagement	3.3	1.05	−0.397 **	−0.433 **	−0.627 **	−0.473 **										
6.	Age [years]	22.4	3.22	−0.020	−0.045	0.068 **	−0.066 **	−0.086 **									
7.	Body Mass Index [kg/m ²]	22.0	3.07	−0.011	0.042	0.069 **	0.026	−0.056 *	−0.193 **								
8.	Perceived Fitness Level	7.2	1.60	−0.281 **	−0.247 **	−0.175 **	−0.189 **	0.156 **	−0.073 **	−0.135 **							
9.	Perceived Healthiness of Diet	4.5	0.89	−0.141 **	−0.148 **	−0.107 **	−0.112 **	0.105 **	0.035	−0.146 **	0.315 **						
10.	Sitting Daily [h]	6.3	3.81	0.086 **	0.136 **	0.063 **	0.098 **	−0.022	0.008	0.034	−0.068 **	−0.048 *					
11.	Physical Activity Total [METS/week]	2762.6	2070.09	−0.060 **	−0.070 **	−0.031	−0.064 **	0.056 *	−0.010	0.010	0.298 **	0.148 **	−0.060 **				
12.	Walking Activity [METS/week]	624.9	708.61	−0.004	−0.027	0.006	−0.010	0.047 *	0.022	−0.004	−0.017	0.056 *	−0.028	0.545 **			
13.	Moderate Activity [METS/week]	617.9	734.92	−0.026	−0.051 *	−0.016	−0.041	0.036	0.049 *	0.008	0.096 **	0.059 **	−0.042	0.678 **	0.303 **		
14.	Vigorous Activity [METS/week]	1591.2	1406.98	−0.073 **	−0.063 **	−0.040	−0.068 **	0.039	−0.052 *	0.013	0.397 **	0.158 **	−0.053 *	0.842 **	0.140 **	0.323 **	
15.	AUDIT-C	3.9	1.96	−0.024	0.010	0.025	0.017	−0.039	−0.042	0.145 **	0.020	−0.141 **	−0.014	0.029	−0.035	−0.014	0.068 **

* Significant at the level of $p \leq 0.05$; ** significant at the level of $p \leq 0.001$.

Table 3. Smoking behavior.

Smoking Behavior	Smokers	Non-Smokers	Occasional Smokers
Men	4.9% (<i>n</i> = 38)	83.5% (<i>n</i> = 654)	11.6% (<i>n</i> = 91)
Women	3.5% (<i>n</i> = 41)	87.5% (<i>n</i> = 1019)	9.0% (<i>n</i> = 105)
Total	4.1% (<i>n</i> = 79)	85.9% (<i>n</i> = 1673)	10.1% (<i>n</i> = 196)

According to the analyses of variances, male students with low, medium and high stress levels differ according to their perceived fitness ($F(2, 780) = 10.585$, $p < 0.001$, $\eta^2 = 0.028$, $n = 778$) and perceived healthiness of their diet ($F(2, 780) = 5.148$, $p = 0.006$, $\eta^2 = 0.013$, $n = 783$). The low-stress group differed significantly from the medium and high group according to GT2 Hochberg's post hoc test for fitness and diet. Regarding that comparison the effect size between low and moderately stressed men was weak ($d = 0.308$) and between low and highly stressed men moderate ($d = 0.475$). Likewise, the difference in diet between low and moderately stressed males was low ($d = 0.198$) and a low to medium effect was observed for low and highly stressed males ($d = 0.381$). Female students could also be distinguished by their stress category in regard to subjective fitness ($F(2, 162) = 47.348$, $p < 0.001$, $\eta^2 = 0.075$, $n = 1165$) and subjective diet ($F(2, 1162) = 17.616$, $\eta^2 = 0.013$, $n = 1165$), the stress groups all differed significantly from another for fitness (low to moderately stressed: $d = 0.413$, low to highly stressed: $d = 0.906$ and moderately to highly stressed: $d = 0.580$) and diet (low to moderately stressed: $d = 0.221$, low to highly stressed: $d = 0.540$ and moderately to highly stressed: $d = 0.382$) in the GT2 Hochberg post hoc testing. Women's stress groups could also be distinguished by sitting hours ($F(2, 1143) = 3.539$, $p = 0.029$, $\eta^2 = 0.006$, $n = 1146$), with the low- and medium-stress groups differing in the post hoc analysis, according to Games-Howell, to a small degree ($d = 0.172$). BMI, daily hours of sitting (men), physical activity or alcohol consumption did not produce significant results in the ANOVA testing.

The Chi-square test could not produce significant results in regard to the distribution of the groups of smokers and the stress category (Table 4).

Table 4. Distribution of smokers in each stress category.

Men	Smoker	Non-Smoker	Occasional Smoker
Low Stress	4.0% (<i>n</i> = 11)	81.7% (<i>n</i> = 227)	14.4% (<i>n</i> = 40)
Medium Stress	5.5% (<i>n</i> = 24)	84.9% (<i>n</i> = 371)	9.6% (<i>n</i> = 42)
High Stress	4.4% (<i>n</i> = 3)	82.4% (<i>n</i> = 56)	13.2% (<i>n</i> = 9)
Total	4.9% (<i>n</i> = 38)	83.5% (<i>n</i> = 654)	11.6% (<i>n</i> = 91)
Women			
Low Stress	2.5% (<i>n</i> = 6)	89.2% (<i>n</i> = 214)	8.3% (<i>n</i> = 20)
Medium Stress	3.2% (<i>n</i> = 24)	87.7% (<i>n</i> = 666)	9.1% (<i>n</i> = 69)
High Stress	6.6% (<i>n</i> = 11)	83.7% (<i>n</i> = 139)	9.6% (<i>n</i> = 16)
Total	3.5% (<i>n</i> = 41)	87.5% (<i>n</i> = 1019)	9.0% (<i>n</i> = 105)

For EE, the groups with frequent symptoms and those without differed in subjective fitness ($t(1955) = 8.922$, $p < 0.001$), subjective healthiness of diet ($t(636.137) = 5.225$, $p < 0.001$), daily sitting hours ($t(582.033) = -3.151$, $p = 0.002$), weekly METS from vigorous activity ($t(1955) = 2.369$, $p = 0.018$) and total weekly METS regardless of the type of activity ($t(1953) = 2.046$, $p = 0.041$) (Table 5).

Table 5. *T*-test EE and health behavior.

	EE Symptoms	N	Mean	Cohen's d	<i>p</i> -Value
BMI	Less than weekly	1507	22.0 ± 2.88		0.295
	Once or more per week	443	22.2 ± 3.64		
Subjective fitness	Less than weekly	1513	7.3 ± 1.50	0.482	<0.001 *
	Once or more per week	444	6.6 ± 1.80		
Subjective healthiness of diet	Less than weekly	1513	4.5 ± 0.84	0.311	<0.001 *
	Once or more per week	444	4.2 ± 1.00		
AUDIT-C	Less than weekly	1356	4.0 ± 1.92		0.604
	Once or more per week	374	3.9 ± 2.12		
Daily sitting hours [h]	Less than weekly	1491	6.2 ± 3.46	0.203	0.002 *
	Once or more per week	440	6.9 ± 4.77		
Weekly METs from vigorous activity	Less than weekly	1513	1560.0 ± 1407.49	0.128	0.018 *
	Once or more per week	444	1380.3 ± 1397.89		
Weekly METs from moderate activity	Less than weekly	1511	634.2 ± 752.36		0.053
	Once or more per week	444	562.2 ± 670.02		
Weekly METs from walking activity	Less than weekly	1511	619.4 ± 689.70		0.552
	Once or more per week	444	643.6 ± 770.03		
Weekly METs from all activity	Less than weekly	1511	2814.5 ± 2084.56	0.129	0.041 *
	Once or more per week	444	2586.0 ± 2012.37		

* Significant at the level of $p \leq 0.05$.

Students with frequent CY had significant higher BMI-values ($t(1948) = -2.091$, $p = 0.37$) than people without frequent symptoms. Subjective fitness was greater in individuals without frequent burnout symptoms ($t(1955) = 5.547$, $p < 0.001$). Students with frequent CY symptoms had more daily sitting hours than their fellow students ($t(1929) = -2.755$, $p = 0.006$) (Table 6).

Table 6. *T*-test CY and health behavior.

	CY Symptoms	N	Mean	Cohen's d	<i>p</i> -Value
BMI	Less than weekly	1808	22.0 ± 3.05	0.182	0.037 *
	Once or more per week	142	22.6 ± 3.32		
Subjective fitness	Less than weekly	1815	7.2 ± 1.56	0.483	<0.001 *
	Once or more per week	142	6.5 ± 1.88		
Subjective healthiness of diet	Less than weekly	1815	4.5 ± 0.87		0.079
	Once or more per week	142	4.3 ± 1.09		
AUDIT-C	Less than weekly	1606	3.94 ± 1.94		0.908
	Once or more per week	124	4.0 ± 2.8		
Daily sitting hours [h]	Less than weekly	1791	6.3 ± 3.83	0.188	0.006 *
	Once or more per week	140	7.2 ± 3.48		
Weekly METs from vigorous activity	Less than weekly	1815	1526.3 ± 1407.54		0.424
	Once or more per week	142	1428.3 ± 1401.57		
Weekly METs from moderate activity	Less than weekly	1813	612.5 ± 729.49		0.250
	Once or more per week	142	686.2 ± 800.67		
Weekly METs from walking activity	Less than weekly	1813	619.3 ± 705.47		0.210
	Once or more per week	142	696.6 ± 746.46		
Weekly METs from all activity	Less than weekly	1813	2758.8 ± 2067.88		0.772
	Once or more per week	142	2811.2 ± 2105.02		

* Significant at the level of $p \leq 0.05$.

For RAE, the datasets have been divided into male and female due to the group differences between the genders. There were no differences in any of the investigated parameters for men (Table 7). Women who had frequent symptoms of RAE, on the other hand, also had a higher BMI ($t(1161) = -2.225$, $p = 0.026$), a poorer subjective fitness

($t(1163) = 4.573$, $p < 0.001$) and perceived their own diet as less healthy ($t(1163) = 3.321$, $p < 0.001$) (Table 8).

Table 7. *T*-test RAE and health behavior—men.

Male	RAE	N	Mean	<i>p</i> -Value
BMI	Less than weekly	683	22.8 ± 3.01	0.156
	Once or more per week	95	23.3 ± 2.94	
Subjective fitness	Less than weekly	688	7.4 ± 1.62	0.235
	Once or more per week	95	7.2 ± 1.65	
Subjective healthiness of diet	Less than weekly	688	4.3 ± 0.90	0.572
	Once or more per week	95	4.3 ± 0.95	
AUDIT-C	Less than weekly	614	4.3 ± 2.04	0.134
	Once or more per week	82	4.9 ± 2.47	
Daily sitting hours [h]	Less than weekly	683	6.7 ± 3.38	0.155
	Once or more per week	93	7.2 ± 3.80	
Weekly METs from vigorous activity	Less than weekly	688	1637.1 ± 1410.24	0.428
	Once or more per week	95	1514.4 ± 1432.04	
Weekly METs from moderate activity	Less than weekly	687	615.0 ± 704.40	0.838
	Once or more per week	95	631.4 ± 909.48	
Weekly METs from walking activity	Less than weekly	687	582.4 ± 612.70	0.470
	Once or more per week	95	631.5 ± 684.26	
Weekly METs from all activity	Less than weekly	687	2834.7 ± 1988.92	0.795
	Once or more per week	95	2777.3 ± 2204.65	

Table 8. *T*-test RAE and health behavior—women.

Female	RAE	N	Mean	Cohen's <i>d</i>	<i>p</i> -Value
BMI	Less than weekly	995	21.4 ± 2.93	0.186	0.026 *
	Once or more per week	168	21.9 ± 3.15		
Subjective fitness	Less than weekly	997	7.2 ± 1.49	0.448	<0.001 *
	Once or more per week	168	6.5 ± 1.87		
Subjective healthiness of diet	Less than weekly	997	4.6 ± 0.84	0.277	<0.001 *
	Once or more per week	168	4.3 ± 0.96		
AUDIT-C	Less than weekly	880	3.6 ± 1.72		0.457
	Once or more per week	148	3.4 ± 1.98		
Daily sitting hours [h]	Less than weekly	981	6.0 ± 4.08		0.127
	Once or more per week	165	6.5 ± 3.69		
Weekly METs from vigorous activity	Less than weekly	997	1479.4.1 ± 1385.61		0.177
	Once or more per week	168	1321.3 ± 1496.15		
Weekly METs from moderate activity	Less than weekly	996	622.1 ± 752.95		0.789
	Once or more per week	168	605.6 ± 652.90		
Weekly METs from walking activity	Less than weekly	996	651.6 ± 769.72		0.884
	Once or more per week	168	642.4 ± 721.45		
Weekly METs from all activity	Less than weekly	996	2754.5 ± 21		0.293
	Once or more per week	168	2569.3.3 ± 2038.52		

* Significant at the level of $p \leq 0.05$.

Cramer's *V*-test revealed that there is a difference in the distribution of female students with and without frequent symptoms of RAE and their smoking behavior (Cramer's $V = 0.074$, $p = 0.043$) (Table 9). There have not been any significant results for the other dimensions of burnout or for men in the RAE dimension.

Table 9. Cross-table RAE and smoking.

Women	RAE	Smoker	Non-Smoker	Occasional Smoker
Less than once a week		3.2% (<i>n</i> = 32)	88.5% (<i>n</i> = 882)	8.3% (<i>n</i> = 83)
Once or more per Week		5.4% (<i>n</i> = 9)	81.5% (<i>n</i> = 137)	13.1% (<i>n</i> = 22)
Total		3.5% (<i>n</i> = 41)	87.5% (<i>n</i> = 1019)	9.0% (<i>n</i> = 105)

Students who were more engaged had a lower BMI ($t(1939) = 1.961$, $p = 0.050$), rated their fitness level ($t(1946) = -5.879$, $p < 0.001$), and healthiness of their diet higher ($t(1946) = -4.740$, $p < 0.001$), and were more active in all activity levels (Table 10). There was no difference in the distribution of engagement and smoking ($p = 0.613$).

Table 10. *T*-test study engagement and health behavior.

	Study Engagement	N	Mean	Cohen's d	<i>p</i> -Value
BMI	Low to medium	1077	22.2 ± 3.20	0.089	0.050 *
	High Engagement	864	21.9 ± 2.88		
Subjective fitness	Low to medium	1080	7.0 ± 1.63	0.272	<0.001 *
	High Engagement	868	7.1 ± 1.52		
Subjective healthiness of diet	Low to medium	1080	4.4 ± 0.90	0.218	<0.001 *
	High Engagement	868	4.6 ± 0.86		
AUDIT-C	Low to medium	957	4.0 ± 2.04		0.166
	High Engagement	767	3.9 ± 1.86		
Daily sitting hours [h]	Low to medium	1066	6.4 ± 4.12		0.706
	High Engagement	856	6.3 ± 3.39		
Weekly METs from vigorous activity	Low to medium	1080	1464.1 ± 1379.65	0.096	0.039 *
	High Engagement	868	1596.7 ± 1441.09		
Weekly METs from moderate activity	Low to medium	1080	582.2 ± 681.67	0.111	0.015 *
	High Engagement	868	664.1 ± 796.58		
Weekly METs from walking activity	Low to medium	1080	592.6 ± 689.76	0.103	0.023 *
	High Engagement	868	666.3 ± 732.52		
Weekly METs from all activity	Low to medium	1080	2638.8 ± 1925.50	0.141	0.003 *
	High Engagement	868	2929.0 ± 2233.00		

* Significant at the level of $p \leq 0.05$.

4. Discussion

The goal of the study was to measure the prevalence of stress, student burnout and study engagement together with students' health behaviors. Associations between these parameters were evaluated while sociodemographic factors were taken into account.

It could be demonstrated that the stress level of students is higher than that of an age-matched reference population [68]. Overall, 12.1% of the students had high stress levels, 61.3% had medium stress levels, while only 26.6% were classified as having low levels of stress. Women had higher stress levels overall. This has been demonstrated in the literature previously [67,75,76]. There are two explanations for this phenomenon, with two different presumptions. The first one is that women are indeed more stressed than men, and that this is due to a higher prevailing anxiety, lesser satisfaction from leisure time and a more intensive rumination of stressful situations [76]. The other presumption is that women are not necessarily more stressed, but they internalize stress rather than men, who tend to externalize stress in form of aggression and impulsive behavior [77], therefore current survey-driven stress measurements may not be as sensitive for men as for women.

Burnout was measured in three dimensions: Emotional Exhaustion (EE), Cynicism (CY) and Reduced Academic Efficacy (RAE). The MBI-SS-Score for EE was 2.6 ± 1.53 , and for CY it was 1.1 ± 1.44 , with no differences between sexes. RAE, however, differed

significantly between sexes. Women showed a significantly higher RAE score of 2.0 ± 1.55 compared to men with a score of 1.8 ± 1.50 ($t(1946) = -2.879$; $p = 0.004$; $d = 0.133$). The literature on sex differences in (student) burnout is inconsistent [78], and the presented results should be taken in light of being a subclinical self-assessment.

All in all, almost one-third (28.1%) of the students show frequent symptoms of burnout in at least one dimension. While EE, with a prevalence of 22.7%, is comparable to German university students, CY's prevalence is lower, while RAE's is higher [28,79].

Not even half (44.5%) of the students were highly engaged, which is slightly less than in a former German-wide investigation [28], but falls into the range of the current literature.

For representation of health habits, BMI, physical activity level, perceived fitness level, perceived healthiness of one's diet and smoking as well as alcohol consumption were chosen as examined factors in addition to sedentary time during the day.

The average BMI for male students was 22.9 ± 3.01 kg/m² and 21.4 ± 2.97 kg/m² for women. For both sexes, more than 80% fell in the normal BMI category of 18.5 to 25.0 kg/m². There were more women who were underweight than men, but more men were overweight or obese than women. In total, the students' BMI average is a little bit lower of that of the age-matched population of Germany [80].

The evaluation of sedentarism among students showed a satisfactory amount of physical activity, with only 7.1% of all students not meeting the minimal requirements [81] and even 41.3% falling into the 'high active' group according to the IPAQ. Average sitting times were 6.3 ± 3.81 h per day. Men underwent more vigorous physical activity, while women walked more. Men estimated their fitness level higher than women, but their diet as less healthy. Male students had more daily sitting hours than female students.

The actual analysis of the associations between stress and the magnitude of the different health habits was performed using an analysis of variances for both male and female students separately. Male as well as female students with low stress levels had a higher perceived fitness level than the moderate- and high-stress groups. This development increased incrementally from the groups of low to high stress, indicating a dose–effect relationship.

Interestingly, the individuals with less frequent burnout symptoms estimated their fitness level as higher than the groups with frequent symptoms. This was true for EE, CY and for female students, and also for RAE. In contrast, the actual PA did not produce significant results for stress as well as for most burnout dimension with the exception of EE, where only vigorous weekly activity was lower in the group with frequent symptoms, with a small effect size of below 0.2 for Cohen's d .

It has been shown in the literature that aerobically trained individuals exhibit lower sympathetic nervous system reactivity in response to physical or psychological stress. In addition, resistance training has been demonstrated to attenuate cardiovascular and hormonal stress responses and improve mental health [82–84]. This can explain why the (subjective) fitness is a valid marker for the perceived stress level of students as well as for burnout symptoms. Vigorous physical activity was significantly and moderately correlated with the perceived fitness level, but it only produced significant results when group testing for EE. In contrast, in a Canadian study by Morgan et al. [58], only low levels of PA have been negatively associated with burnout. The inconsistent relationship of PA and burnout/stress may be due to their bidirectional relationship. So, while PA can prevent and counteract stress [85,86], it is also often used as a coping strategy in times of increased stress, which is particularly true for individuals who exercise regularly [87]. Therefore, cross-sectional analyses may have trouble, defining the relationship of PA and stress. Moreover, the fitness level is the long-term accumulation of a healthy lifestyle. It also often reflects the PA level of an individual's parenting home and the amount and satisfaction with sports in primary and secondary school [88]. Fitness could therefore be a

better marker in the sense that it can be used as a more distinct indicator of the accumulated stress-resilience due to a long-term active lifestyle and regular training in past (and present), while the IPAQ only asks about PA within the last seven days. While research supports that all types of PA can be beneficial to physical and psychological health [89], we need more research to examine the relationships between total PA and burnout. However, we will point out that we evaluated the perceived instead of objective fitness. This variable might be skewed by the self-esteem and personality traits such as optimism/pessimism, which also influence resilience independently [90].

In the same sense, a sedentary lifestyle was more often seen in individuals with higher scores on the PSS-10, and those who had frequent symptoms of EE and CY. Sedentary behavior is linked to both physical and psychological ill health, which is also true for college students [91,92]. Stress and sedentary behavior have been positively linked in some studies before [91,93] but never in the subpopulation of German university students so far. Interestingly, Morgan et al. [58] found the opposite to be the case and concluded that (medical) students who were not able to allot enough time for restful activities may be at an increased risk of burnout. The right balance might be crucial for health and performance among students.

The perceived healthiness of the individuals' diet followed a similar pattern: Students who were less stressed and showed fewer symptoms of burnout (except for the dimension of CY) tended to report that their diet was healthier. A higher BMI, which is linked to diet (but not to diet alone), was associated with more frequent symptoms of CY and RAE among women. Unhealthy dietary habits are not only common among university students but have also been linked to stress, anxiety, depression and insomnia in former studies [94]. Related to this, an increase in BMI can be observed in many students, especially during their first year of university [95,96]. Shifts in dietary habits are associated with stress and, in conjunction with the lack of adequate PA, may lead to the weight gain seen in many students [97]. Also, burnout has been linked to unhealthy diets before [98], but, to our knowledge, never in this subpopulation of students.

Interestingly, the only significant result for alcohol or tobacco consumption could be found for women in the dimension of RAE. There, we found significantly more smokers and occasional smokers in the group of women with frequent RAE symptoms. We expected to see this trend more often, as mental ill health, including stress and depression, is often linked to a more (excessive) use of alcohol and tobacco [99] which was not present in our sample. On the other hand, alcohol facilitates social connecting among university students and reduces anxiety and supports successful transition within a new peer group and has been labeled as 'a successful tool for hastening social connectedness' by Brown and Murphy [100], therefore providing at least short-term social benefits for students.

For study engagement, similar but counter-directed results have been found, which is not surprising given the close inverse relationship between burnout and engagement [12].

Engaged students have been found to have a lower BMI and a higher estimate of their own fitness level and the healthiness of their diet. In contrast to stress and burnout, PA could be identified as a more relevant association with engagement. Only little is known about study engagement and health behavior and to our knowledge PA has not been analyzed in this context at all to this day. However, mechanisms that link PA and engagement are plausible. Rozanski [101], e.g., explored the role of health habits and psychological determinants on 'vitality', a pleasing sensation of feeling energetic, which is a core attribute of study engagement. PA has been shown to be connected to attention and concentration, a key element of the absorption dimension within study engagement [102]. On top of that, it has already been demonstrated how healthy habits including PA are able to reduce job stress [103]. Lubans et al. [104] could also link self-regulation and PA, while

Zhang et al. [105] was able to show that engaged students seem to exhibit self-regulation, which might represent a possibly link between PA and study engagement in addition to the general benefits PA offers for mental health and resilience [106,107].

5. Conclusions

The study showed that stress, burnout and low study engagement are prevalent among students in a sample of different subject fields. Health habits such as an active lifestyle, a healthy diet and fitness levels are associated with all three factors. While fitness and diet were most often associated, the role of BMI and PA was not as pronounced as expected. To understand and prevent the cumulative effects of study demands and to identify subgroups and according interventions, the analysis of health behavior is important, especially for the creation of successful behavioral prevention. The study also contributes to the study–demands–resources model [33]. The primary focus of the SDR has, to date, mostly been structural elements and psychological factors. Our study also shows that personal habits may play an important role for student burnout and study engagement as psychological constructs. This study therefore broadens the perspective by highlighting a possible role of personal habits in influencing student burnout and study engagement. Our findings suggest that in order to effectively address student burnout and foster study engagement, interventions must go beyond structural and psychological support. They should also incorporate strategies that promote personal healthy habits. This calls for a more holistic approach in the SDR that includes behavioral prevention and self-regulation practices as key components. However, future research needs to confirm a causal relationship using longitudinal or intervention design.

Non-binary individuals have been excluded due to the small sample size. Based on the descriptive data, we see the need for more inclusive research to further investigate this marginalized target group.

Strength and Limitations

The study was—to our knowledge—the first to evaluate the effect of different intensities of PA on a healthy students' collective of different subject fields and also the first to analyze the effect of the included health habits on study engagement. The inclusion of students of all subject fields is one of the strengths of this study, because often—and maybe erroneously [29]—the focus has been placed on medical students instead of on the whole student collective.

On the other hand, this study used the students' own perception of their diet and fitness level instead of objective information for these variables. It is likely that students will misjudge their own fitness level. Likewise, the self-reported estimation of one's diet does not allow for objective markers on what qualifies a diet as healthy. Also, students who know less about healthy eating might misjudge their own diet even more. Previous research suggested that an individual's definition of healthy eating can often be guided by one or few predominant themes (e.g., eating in order to control weight, presence of specific micro- or macronutrients, etc.). However, a uniform definition of a healthy diet does not exist, yet. It is remarkable that despite that fact, the perceived healthiness of the diet is a consistent marker for stress, burnout and engagement in this study. It is, however, plausible that a higher estimate of an individual's diet and fitness could also be an expression of an individual's self-perception, which is also linked to self-efficacy and mental health [108], therefore leaving room for a possible bias. More objective data on the physical fitness of students should be used to confirm the above findings.

On top of that, most of these factors are described to be in a reciprocal relationship to mental health as they seem to increase the resilience on the one hand but serve as (negative)

coping strategies on the other. Longitudinal or experimental studies are needed to confirm a causal relationship.

Furthermore, the large sample size increased statistical power, leading to some statistically significant results despite very small effect sizes ($d < 0.2$). These results may have limited practical validity in the context of a singular study.

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Abbreviations

The following abbreviations are used in this manuscript:

EE	Emotional Exhaustion
CY	Cynicism
RAE	Reduced Academic Efficacy
SDR	Study–Demands–Resources (Model)
PA	Physical Activity
AUDIT-C	Alcohol Use Disorders Identification Test
UWES-S-9	Utrecht Work Engagement Scale Students-9 Items
MBI-SS	Maslach Burnout Inventory Student Shortform
PSS-10	Perceived Stress Scale-10 Items
IPAQ	International Physical Activity Questionnaire
MET	Metabolic Equivalent

References

1. Edú-Valsania, S.; Lagufa, A.; Moriano, J.A. Burnout: A Review of Theory and Measurement. *Int. J. Environ. Res. Public Health* **2022**, *19*, 1780. [[CrossRef](#)] [[PubMed](#)]
2. Freudenberg, H.J. Staff burn-out. *J. Soc. Issues* **1974**, *30*, 159–165. [[CrossRef](#)]
3. Han, S.; Shanafelt, T.D.; Sinsky, C.A.; Awad, K.M.; Dyrbye, L.N.; Fiscus, L.C.; Trockel, M.; Goh, J. Estimating the Attributable Cost of Physician Burnout in the United States. *Ann. Intern. Med.* **2019**, *170*, 784–790. [[CrossRef](#)] [[PubMed](#)]
4. Dyrbye, L.N.; Thomas, M.R.; Massie, F.S.; Power, D.V.; Eacker, A.; Harper, W.; Durning, S.; Moutier, C.; Szydlo, D.W.; Novotny, P.J.; et al. Burnout and suicidal ideation among U.S. medical students. *Ann. Intern. Med.* **2008**, *149*, 334–341. [[CrossRef](#)]
5. Edwards, D.; Burnard, P.; Bennett, K.; Hebden, U. A longitudinal study of stress and self-esteem in student nurses. *Nurse Educ. Today* **2010**, *30*, 78–84. [[CrossRef](#)]
6. Moneta, G.B. Need for achievement, burnout, and intention to leave: Testing an occupational model in educational settings. *Personal. Individ. Differ.* **2011**, *50*, 274–278. [[CrossRef](#)]

7. Maslach, C. Job burnout: New directions in research and intervention. *Curr. Dir. Psychol. Sci.* **2003**, *12*, 189–192. [\[CrossRef\]](#)
8. Bullock, A.; Barnes, E.; Morris, Z.S.; Fairbank, J.; de Pury, J.; Howell, R.; Denman, S. *Getting the Most out of Knowledge and Innovation Transfer Agents in Health Care: A Qualitative Study*; Health Services and Delivery Research: Southampton, UK, 2016.
9. Hu, Q.; Schaufeli, W.B. The factorial validity of the Maslach Burnout Inventory-Student Survey in China. *Psychol. Rep.* **2009**, *105*, 394–408. [\[CrossRef\]](#)
10. Lindfors, P.; Minkkinen, J.; Rimpelä, A.; Hotulainen, R. Family and school social capital, school burnout and academic achievement: A multilevel longitudinal analysis among Finnish pupils. *Int. J. Adolesc. Youth* **2018**, *23*, 368–381. [\[CrossRef\]](#)
11. Mostert, K.; Pienaar, J.; Gauche, C.; Jackson, L. Burnout and engagement in university students: A psychometric analysis of the MBI-SS and UWES-S. *S. Afr. J. High. Educ.* **2007**, *21*, 147–162. [\[CrossRef\]](#)
12. Schaufeli, W.B.; Martinez, I.M.; Pinto, A.M.; Salanova, M.; Bakker, A.B. Burnout and engagement in university students: A cross-national study. *J. Cross-Cult. Psychol.* **2002**, *33*, 464–481. [\[CrossRef\]](#)
13. Schaufeli, W.B.; Salanova, M.; González-Romá, V.; Bakker, A.B. The measurement of engagement and burnout: A two sample confirmatory factor analytic approach. *J. Happiness Stud.* **2002**, *3*, 71–92. [\[CrossRef\]](#)
14. Tuominen-Soini, H.; Salmela-Aro, K.; Niemivirta, M. Achievement goal orientations and academic well-being across the transition to upper secondary education. *Learn. Individ. Differ.* **2012**, *22*, 290–305. [\[CrossRef\]](#)
15. Christian, M.S.; Slaughter, J.E. (Eds.) Work engagement: A meta-analytic review and directions for research in an emerging area. In *Academy of Management Proceedings*; Academy of Management: Briarcliff Manor, NY, USA, 2007; p. 10510.
16. Olson, N.; Oberhoffer-Fritz, R.; Reiner, B.; Schulz, T. Study related factors associated with study engagement and student burnout among German university students. *Front. Public Health* **2023**, *11*, 1168264. [\[CrossRef\]](#)
17. Heaven, P. *Adolescent Health: The Role of Individual Differences*; Psychology Press: London, UK, 2002.
18. Murray, K.M.; Byrne, D.G.; Rieger, E. Investigating adolescent stress and body image. *J. Adolesc.* **2011**, *34*, 269–278. [\[CrossRef\]](#)
19. Seifert, T.A.; Bowman, N.A.; Wolniak, G.C.; Rockenbach, A.N.; Mayhew, M.J. Ten challenges and recommendations for advancing research on the effects of college on students. *AERA Open* **2017**, *3*, 2332858417701683. [\[CrossRef\]](#)
20. Arnett, J.J. Emerging adulthood. A theory of development from the late teens through the twenties. *Am. Psychol.* **2000**, *55*, 469–480. [\[CrossRef\]](#) [\[PubMed\]](#)
21. Mortier, P.; Auerbach, R.P.; Alonso, J.; Bantjes, J.; Benjet, C.; Cuijpers, P.; Ebert, D.D.; Green, J.G.; Hasking, P.; Nock, M.K.; et al. Suicidal thoughts and behaviors among first-year college students: Results from the WMH-ICS project. *J. Am. Acad. Child Adolesc. Psychiatry* **2018**, *57*, 263–273.e1. [\[CrossRef\]](#)
22. Almeida, G.d.C.; Souza HRd Almeida PCd Almeida, B.d.C.; Almeida, G.H. The prevalence of burnout syndrome in medical students. *Arch. Clin. Psychiatry* **2016**, *43*, 6–10. [\[CrossRef\]](#)
23. Boni, R.; Paiva, C.E.; de Oliveira, M.A.; Lucchetti, G.; Fregnani, J.; Paiva, B.S.R. Burnout among medical students during the first years of undergraduate school: Prevalence and associated factors. *PLoS ONE* **2018**, *13*, e0191746. [\[CrossRef\]](#)
24. Bullock, G.; Kraft, L.; Amsden, K.; Gore, W.; Prengle, B.; Wimsatt, J.; Prengle, R.; Ledbetter, L.; Covington, K.; Goode, A. The prevalence and effect of burnout on graduate healthcare students. *Can. Med. Educ. J.* **2017**, *8*, e90–e108. [\[CrossRef\]](#) [\[PubMed\]](#)
25. Ishak, W.; Nikraves, R.; Lederer, S.; Perry, R.; Ogunyemi, D.; Bernstein, C. Burnout in medical students: A systematic review. *Clin. Teach.* **2013**, *10*, 242–245. [\[CrossRef\]](#) [\[PubMed\]](#)
26. Li, Y.; Cao, L.; Liu, J.; Zhang, T.; Yang, Y.; Shi, W.; Wei, Y. The prevalence and associated factors of burnout among undergraduates in a university. *Medicine* **2021**, *100*, e26589. [\[CrossRef\]](#)
27. Braun, M.; Schönfeldt-Lecuona, C.; Kessler, H.; Beck, J.; Beschoner, P.; Freudenmann, R.W. Burnout, Depression und Substanzgebrauch bei deutschen Psychiatern und Nervenärzten. *Nervenheilkunde* **2008**, *27*, 800–804.
28. Grützner, J.; Gussy, B.; Lesener, T.; Sudheimer, S.; Willige, J. *Gesundheit Studierender in Deutschland 2017*; Ein Kooperationsprojekt zwischen dem Deutschen Zentrum für Hochschul- und Wissenschaftsforschung; Freien Universität Berlin: Berlin, Germany, 2018; pp. 1–168.
29. Olson, N.; Oberhoffer-Fritz, R.; Reiner, B.; Schulz, T. Stress, student burnout and study engagement—A cross-sectional comparison of university students of different academic subjects. *BMC Psychol.* **2025**, *13*, 293. [\[CrossRef\]](#)
30. Salmela-Aro, K.; Read, S. Study engagement and burnout profiles among Finnish higher education students. *Burn. Res.* **2017**, *7*, 21–28. [\[CrossRef\]](#)
31. Loscalzo, Y.; Giannini, M. Study engagement in Italian university students: A confirmatory factor analysis of the Utrecht Work Engagement Scale—Student version. *Soc. Indic. Res.* **2019**, *142*, 845–854. [\[CrossRef\]](#)
32. Robins, T.G.; Roberts, R.M.; Sarris, A. Burnout and engagement in health profession students: The relationships between study demands, study resources and personal resources. *Australas. J. Organ. Psychol.* **2015**, *8*, e1. [\[CrossRef\]](#)
33. Gussy, B.; Wörfel, F.; Lohmann, K. Erschöpfung und Engagement im Studium: Eine Anwendung des Job Demands-Resources Modells. *Z. Gesundheitspsychol.* **2016**, *24*, 41–53. [\[CrossRef\]](#)
34. Lizarte Simón, E.J.; Gijón Puerta, J.; Galván Malagón, M.C.; Khaled Gijón, M. Influence of Self-Efficacy, Anxiety and Psychological Well-Being on Academic Engagement During University Education. *Educ. Sci.* **2024**, *14*, 1367. [\[CrossRef\]](#)

35. Heuse, S.; Risius, U.-M. Stress bei Studierenden mit und ohne Nebenjob. *Prävention Und Gesundheitsförderung* **2022**, *17*, 379–384. [\[CrossRef\]](#)
36. Liu, H.; Yansane, A.I.; Zhang, Y.; Fu, H.; Hong, N.; Kalendarian, E. Burnout and study engagement among medical students at Sun Yat-sen University, China: A cross-sectional study. *Medicine* **2018**, *97*, e0326. [\[CrossRef\]](#) [\[PubMed\]](#)
37. Salmela-Aro, K.; Upadaya, K. School burnout and engagement in the context of demands-resources model. *Br. J. Educ. Psychol.* **2014**, *84 Pt 1*, 137–151. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Friedman, S.M. Lifestyle (Medicine) and Healthy Aging. *Clin. Geriatr. Med.* **2020**, *36*, 645–653. [\[CrossRef\]](#)
39. Walsh, R. Lifestyle and mental health. *Am. Psychol.* **2011**, *66*, 579–592. [\[CrossRef\]](#)
40. Puente-Hidalgo, S.; Prada-García, C.; Benítez-Andrades, J.A.; Fernández-Martínez, E. Promotion of Healthy Habits in University Students: Literature Review. *Healthcare* **2024**, *12*, 993. [\[CrossRef\]](#)
41. Freedman, M.R.; Waldrop, J. Freshman orientation sessions can teach incoming students about healthful lifestyles. *J. Nutr. Educ. Behav.* **2011**, *43*, 69–70. [\[CrossRef\]](#) [\[PubMed\]](#)
42. Ball, S.; Bax, A. Self-care in medical education: Effectiveness of health-habits interventions for first-year medical students. *Acad. Med.* **2002**, *77*, 911–917. [\[CrossRef\]](#)
43. Buckworth, J.; Nigg, C. Physical activity, exercise, and sedentary behavior in college students. *J. Am. Coll. Health* **2004**, *53*, 28–34. [\[CrossRef\]](#)
44. Irwin, J.D. Prevalence of university students' sufficient physical activity: A systematic review. *Percept. Mot. Ski.* **2004**, *98 Pt 1*, 927–943. [\[CrossRef\]](#)
45. Keating, X.D.; Guan, J.; Piñero, J.C.; Bridges, D.M. A meta-analysis of college students' physical activity behaviors. *J. Am. Coll. Health* **2005**, *54*, 116–125. [\[CrossRef\]](#) [\[PubMed\]](#)
46. Salzwedel, A.; Rabe, S.; Zahn, T.; Neuwirth, J.; Eichler, S.; Haubold, K.; Wachholz, A.; Relbls, R.; Voller, H. Nutzerinteresse an mobilen digitalen Assistenzsystemen zur Förderung körperlicher Aktivität bei Studierenden und Mitarbeitern einer Universität. *Das Gesundheitswesen* **2018**, *80*, 1023–1025. [\[CrossRef\]](#) [\[PubMed\]](#)
47. Sebena, R.; El Ansari, W.; Stock, C.; Orosova, O.; Mikolajczyk, R.T. Are perceived stress, depressive symptoms and religiosity associated with alcohol consumption? A survey of freshmen university students across five European countries. *Subst. Abuse. Treat. Prev. Policy* **2012**, *7*, 21. [\[CrossRef\]](#) [\[PubMed\]](#)
48. Cluskey, M.; Grobe, D. College weight gain and behavior transitions: Male and female differences. *J. Am. Diet. Assoc.* **2009**, *109*, 325–329. [\[CrossRef\]](#)
49. Gores, S.E. Addressing nutritional issues in the college-aged client: Strategies for the nurse practitioner. *J. Am. Acad. Nurse Pract.* **2008**, *20*, 5–10. [\[CrossRef\]](#)
50. Ingram, R.E.; Price, J.M. *Vulnerability to Psychopathology: Risk Across the Lifespan*; Guilford Press: New York, NY, USA, 2010.
51. Alhajj, M.N.; Khader, Y.; Murad, A.H.; Celebic, A.; Halboub, E.; Márquez, J.R.; Macizo, C.C.; Khan, S.; Basnet, B.B.; Makzoumé, J.E.; et al. Perceived sources of stress amongst dental students: A multicountry study. *Eur. J. Dent. Educ.* **2018**, *22*, 258–271. [\[CrossRef\]](#)
52. Park, J.H.; Moon, J.H.; Kim, H.J.; Kong, M.H.; Oh, Y.H. Sedentary Lifestyle: Overview of Updated Evidence of Potential Health Risks. *Korean J. Fam. Med.* **2020**, *41*, 365–373. [\[CrossRef\]](#)
53. Flegal, K.M.; Kit, B.K.; Orpana, H.; Graubard, B.I. Association of all-cause mortality with overweight and obesity using standard body mass index categories: A systematic review and meta-analysis. *JAMA* **2013**, *309*, 71–82. [\[CrossRef\]](#) [\[PubMed\]](#)
54. Głowska, D.; Guzek, D.; Groele, B.; Gutkowska, K. Fruit and Vegetable Intake and Mental Health in Adults: A Systematic Review. *Nutrients* **2020**, *12*, 115. [\[CrossRef\]](#)
55. Lane, M.M.; Gamage, E.; Travica, N.; Dissanayaka, T.; Ashtree, D.N.; Gauci, S.; Lotfaliany, M.; O'neil, A.; Jacka, F.N.; Marx, W. Ultra-Processed Food Consumption and Mental Health: A Systematic Review and Meta-Analysis of Observational Studies. *Nutrients* **2022**, *14*, 2568. [\[CrossRef\]](#)
56. Ventriglio, A.; Sancassiani, F.; Contu, M.P.; Latorre, M.; Di Salvatore, M.; Fornaro, M.; Bhugra, D. Mediterranean Diet and its Benefits on Health and Mental Health: A Literature Review. *Clin. Pr. Epidemiol. Ment. Health* **2020**, *16* (Suppl. 1), 156–164. [\[CrossRef\]](#) [\[PubMed\]](#)
57. Warburton, D.E.R.; Bredin, S.S.D. Health benefits of physical activity: A systematic review of current systematic reviews. *Curr. Opin. Cardiol.* **2017**, *32*, 541–556. [\[CrossRef\]](#)
58. Morgan, T.L.; McFadden, T.; Fortier, M.S.; Sweet, S.N.; Tomasone, J.R. Do physical activity intensity and sedentary behaviour relate to burnout among medical students? Insight from two Canadian medical schools. *Can. Med. Educ. J.* **2024**, *15*, 54–63. [\[CrossRef\]](#) [\[PubMed\]](#)
59. Maslach, C.; Jackson, S.E. The measurement of experienced burnout. *J. Organ. Behav.* **1981**, *2*, 99–113. [\[CrossRef\]](#)
60. Shadid, A.; Shadid, A.M.; Shadid, A.; Almutairi, F.E.; Almotairi, K.E.; Aldarwish, T.; Alzamil, O.; Alkholaiwi, F.; Khan, S.-U.-D. Stress, Burnout, and Associated Risk Factors in Medical Students. *Cureus* **2020**, *12*, e6633. [\[CrossRef\]](#)

61. Gumz, A.; Erices, R.; Brähler, E.; Zenger, M. Faktorstruktur und Gütekriterien der deutschen Übersetzung des Maslach-Burnout-Inventars für Studierende von Schaufeli et al. (MBI-SS). *PPmP Psychother. Psychosom. Med. Psychol.* **2013**, *63*, 77–84. [\[CrossRef\]](#) [\[PubMed\]](#)
62. Morales-Rodríguez, F.M.; Pérez-Mármol, J.M.; Brown, T. Education Burnout and Engagement in Occupational Therapy Undergraduate Students and Its Associated Factors. *Front. Psychol.* **2019**, *10*, 2889. [\[CrossRef\]](#) [\[PubMed\]](#)
63. Seppälä, P.; Mauno, S.; Feldt, T.; Hakanen, J.; Kinnunen, U.; Tolvanen, A.; Schaufeli, W. The construct validity of the Utrecht Work Engagement Scale: Multisample and longitudinal evidence. *J. Happiness Stud.* **2009**, *10*, 459–481. [\[CrossRef\]](#)
64. Gusy, B.; Lesener, T.; Wolter, C. Measuring well-being with the Utrecht work engagement scale–student form. *Eur. J. Health Psychol.* **2019**, *26*, 31–38. [\[CrossRef\]](#)
65. Dias, J.C.R.; Silva, W.R.; Maroco, J.; Campos, J.A.D.B. Perceived stress scale applied to college students: Validation study. *Psychol. Community Health* **2015**, *4*, 1–13. [\[CrossRef\]](#)
66. Klein, E.M.; Brähler, E.; Dreier, M.; Reinecke, L.; Müller, K.W.; Schmutzer, G.; Wölfling, K.; Beutel, M.E. The German version of the Perceived Stress Scale–psychometric characteristics in a representative German community sample. *BMC Psychiatry* **2016**, *16*, 1–10. [\[CrossRef\]](#) [\[PubMed\]](#)
67. Cavallo, P.; Carpinelli, L.; Savarese, G. Perceived stress and bruxism in university students. *BMC Res. Notes* **2016**, *9*, 514. [\[CrossRef\]](#) [\[PubMed\]](#)
68. Cohen, S.; Kamarck, T.; Mermelstein, R. Perceived stress scale. *Meas. Stress: A Guide Health Soc. Sci.* **1994**, *10*, 1–2.
69. Swaminathan, A.; Viswanathan, S.; Gnanadurai, T.; Ayyavoo, S.; Manickam, T. Perceived stress and sources of stress among first-year medical undergraduate students in a private medical college Tamil Nadu. *Natl. J. Physiol. Pharm. Pharmacol.* **1970**, *6*, 9–14. [\[CrossRef\]](#)
70. Thangaraj, S.; D’souza, L. Prevalence of Stress Levels Among First Year Medical Undergraduate Students. *Int. J. Interdiscip. Multidiscip. Stud.* **2014**, *1*, 176–181.
71. Craig, C.L.; Marshall, A.L.; Sjöström, M.; Bauman, A.E.; Booth, M.L.; Ainsworth, B.E.; Pratt, M.; Ekelund, U.L.; Yngve, A.; Sallis, J.F.; et al. International physical activity questionnaire: 12-country reliability and validity. *Med. Sci. Sports Exerc.* **2003**, *35*, 1381–1395. [\[CrossRef\]](#)
72. IPAQ Scoring protocol for the International Physical Activity Questionnaire (IPAQ). 2022. Available online: <https://sites.google.com/view/ipaq/score> (accessed on 30 May 2025).
73. Bradley, K.A.; DeBenedetti, A.F.; Volk, R.J.; Williams, E.C.; Frank, D.; Kivlahan, D.R. AUDIT-C as a brief screen for alcohol misuse in primary care. *Alcohol. Clin. Exp. Res.* **2007**, *31*, 1208–1217. [\[CrossRef\]](#)
74. Balducci, C.; Fraccaroli, F.; Schaufeli, W.B. Psychometric properties of the Italian version of the Utrecht Work Engagement Scale (UWES-9). *Eur. J. Psychol. Assess.* **2010**, *26*, 143–149. [\[CrossRef\]](#)
75. Beiter, R.; Nash, R.; McCrady, M.; Rhoades, D.; Linscomb, M.; Clarahan, M.; Sammut, S. The prevalence and correlates of depression, anxiety, and stress in a sample of college students. *J. Affect. Disord.* **2015**, *173*, 90–96. [\[CrossRef\]](#)
76. Nechita, F.; Nechita, D.; Pîrlog, M.C.; Rogoveanu, I. Stress in medical students. *Rom. J. Morphol. Embryol.* **2014**, *55* (Suppl. 3), 1263–1266.
77. Boyd, A.; Van de Velde, S.; Vilagut, G.; de Graaf, R.; O’Neill, S.; Florescu, S.; Alonso, J.; Kovess-Masfety, V. Gender differences in mental disorders and suicidality in Europe: Results from a large cross-sectional population-based study. *J. Affect. Disord.* **2015**, *173*, 245–254. [\[CrossRef\]](#) [\[PubMed\]](#)
78. Purvanova, R.K.; Muros, J.P. Gender differences in burnout: A meta-analysis. *J. Vocat. Behav.* **2010**, *77*, 168–185. [\[CrossRef\]](#)
79. Wörfel, F.; Gusy, B.; Lohman, K.; Kleiber, D. Validierung der deutschen Kurzversion des Maslach-Burnout-Inventars für Studierende (MBI-SS KV). *Z. Für Gesundheitspsychol.* **2016**, *23*, 191–196. [\[CrossRef\]](#)
80. Bundesamt, S. Anteile an der Deutschen Erwachsenenbevölkerung nach Körpermaßen (BMI)* und Altersgruppen im Jahr 2021. 2023. Available online: <https://de.statista.com/statistik/daten/studie/256578/umfrage/bevoelkerungsanteile-in-deutschland-nach-koerpermassen-bmi-und-altersgruppen/> (accessed on 30 May 2025).
81. WHO. *Physical Activity Strategy for the WHO European Region 2016–2025*; WHO: Geneva, Switzerland, 2016.
82. Huang, C.J.; Webb, H.E.; Zourdos, M.C.; Acevedo, E.O. Cardiovascular reactivity, stress, and physical activity. *Front. Physiol.* **2013**, *4*, 314. [\[CrossRef\]](#)
83. Rimmele, U.; Seiler, R.; Marti, B.; Wirtz, P.H.; Ehlert, U.; Heinrichs, M. The level of physical activity affects adrenal and cardiovascular reactivity to psychosocial stress. *Psychoneuroendocrinology* **2009**, *34*, 190–198. [\[CrossRef\]](#) [\[PubMed\]](#)
84. Spalding, T.W.; Jeffers, L.S.; Porges, S.W.; Hatfield, B.D. Vagal and cardiac reactivity to psychological stressors in trained and untrained men. *Med. Sci. Sports Exerc.* **2000**, *32*, 581–591. [\[CrossRef\]](#) [\[PubMed\]](#)
85. Salmon, P. Effects of physical exercise on anxiety, depression, and sensitivity to stress: A unifying theory. *Clin. Psychol. Rev.* **2001**, *21*, 33–61. [\[CrossRef\]](#)

86. Schnohr, P.; Kristensen, T.S.; Prescott, E.; Scharling, H. Stress and life dissatisfaction are inversely associated with jogging and other types of physical activity in leisure time—The Copenhagen City Heart Study. *Scand. J. Med. Sci. Sports* **2005**, *15*, 107–112. [\[CrossRef\]](#)
87. Stults-Kolehmainen, M.A.; Sinha, R. The effects of stress on physical activity and exercise. *Sports Med.* **2014**, *44*, 81–121. [\[CrossRef\]](#)
88. Carballo-Fazanes, A.; Rico-Díaz, J.; Barcala-Furelos, R.; Rey, E.; Rodríguez-Fernández, J.E.; Varela-Casal, C.; Abelairas-Gómez, C. Physical Activity Habits and Determinants, Sedentary Behaviour and Lifestyle in University Students. *Int. J. Environ. Res. Public Health* **2020**, *17*, 3272. [\[CrossRef\]](#)
89. Kikuchi, H.; Inoue, S.; Lee, I.M.; Odagiri, Y.; Sawada, N.; Inoue, M.; Tsugane, S. Impact of Moderate-Intensity and Vigorous-Intensity Physical Activity on Mortality. *Med. Sci. Sports Exerc.* **2018**, *50*, 715–721. [\[CrossRef\]](#)
90. Malak-Akgün, B.; Üzar-Özçetin, Y.S.; Aydin, A. Association between resilience, self-esteem and suicide probability among university students in Turkey. *Perspect. Psychiatr. Care* **2022**, *58*, 2524–2533. [\[CrossRef\]](#)
91. Gao, X.L.; Zhang, J.H.; Yang, Y.; Cao, Z.B. Sedentary behavior, screen time and mental health of college students: A Meta-analysis. *Zhonghua Liu Xing Bing Xue Za Zhi* **2023**, *44*, 477–485. [\[PubMed\]](#)
92. Herbert, C. Enhancing Mental Health, Well-Being and Active Lifestyles of University Students by Means of Physical Activity and Exercise Research Programs. *Front. Public Health* **2022**, *10*, 849093. [\[CrossRef\]](#)
93. Lee, E.; Kim, Y. Effect of university students' sedentary behavior on stress, anxiety, and depression. *Perspect. Psychiatr. Care* **2019**, *55*, 164–169. [\[CrossRef\]](#) [\[PubMed\]](#)
94. Ramón-Arbués, E.; Martínez Abadía, B.; Granada López, J.M.; Echániz Serrano, E.; Pellicer García, B.; Juárez Vela, R.; Portillo, S.G.; Guinoa, M.S. Eating behavior and relationships with stress, anxiety, depression and insomnia in university students. *Nutr. Hosp.* **2019**, *36*, 1339–1345. [\[CrossRef\]](#) [\[PubMed\]](#)
95. Holm-Denoma, J.M.; Joiner, T.E.; Vohs, K.D.; Heatherton, T.F. The “freshman fifteen” (the “freshman five” actually): Predictors and possible explanations. *Health Psychol.* **2008**, *27* (Suppl. S1), S3–S9. [\[CrossRef\]](#)
96. Vella-Zarb, R.A.; Elgar, F.J. Predicting the ‘freshman 15’: Environmental and psychological predictors of weight gain in first-year university students. *Health Educ. J.* **2010**, *69*, 321–332. [\[CrossRef\]](#)
97. Choi, J. Impact of Stress Levels on Eating Behaviors among College Students. *Nutrients* **2020**, *12*, 1241. [\[CrossRef\]](#)
98. Penttinen, M.A.; Virtanen, J.; Laaksonen, M.; Erkkola, M.; Vepsäläinen, H.; Kautiainen, H.; Korhonen, P. The Association between Healthy Diet and Burnout Symptoms among Finnish Municipal Employees. *Nutrients* **2021**, *13*, 2393. [\[CrossRef\]](#)
99. Chow, M.S.C.; Poon, S.H.L.; Lui, K.L.; Chan, C.C.Y.; Lam, W.W.T. Alcohol Consumption and Depression Among University Students and Their Perception of Alcohol Use. *East Asian Arch. Psychiatry* **2021**, *31*, 87–96. [\[CrossRef\]](#) [\[PubMed\]](#)
100. Brown, R.; Murphy, S. Alcohol and social connectedness for new residential university students: Implications for alcohol harm reduction. *J. Furth. High. Educ.* **2020**, *44*, 216–230. [\[CrossRef\]](#) [\[PubMed\]](#)
101. Rozanski, A. The pursuit of health: A vitality based perspective. *Prog. Cardiovasc. Dis.* **2023**, *77*, 14–24. [\[CrossRef\]](#)
102. Reigal, R.E.; Moral-Campillo, L.; de Mier, R.J.; Morillo-Baro, J.P.; Morales-Sánchez, V.; Pastrana, J.L.; Hernández-Mendo, A. Physical Fitness Level Is Related to Attention and Concentration in Adolescents. *Front. Psychol.* **2020**, *11*, 110. [\[CrossRef\]](#)
103. Linn, L.S.; Cope, D.; Leake, B.; Yager, J. Health habits and coping behaviors among practicing physicians. *West. J. Med.* **1986**, *144*, 484. [\[PubMed\]](#)
104. Lubans, D.; Richards, J.; Hillman, C.; Faulkner, G.; Beauchamp, M.; Nilsson, M.; Kelly, P.; Smith, J.; Raine, L.; Biddle, S. Physical activity for cognitive and mental health in youth: A systematic review of mechanisms. *Pediatrics* **2016**, *138*, e20161642. [\[CrossRef\]](#)
105. Zhang, S.; Shi, R.; Yun, L.; Li, X.; Wang, Y.; He, H.; Miao, D. Self-regulation and study-related health outcomes: A structural equation model of regulatory mode orientations, academic burnout and engagement among university students. *Soc. Indic. Res.* **2015**, *123*, 585–599. [\[CrossRef\]](#)
106. Peluso, M.A.; Guerra de Andrade, L.H. Physical activity and mental health: The association between exercise and mood. *Clinics* **2005**, *60*, 61–70. [\[CrossRef\]](#)
107. Raglin, J.S. Exercise and mental health. Beneficial and detrimental effects. *Sports Med.* **1990**, *9*, 323–329. [\[CrossRef\]](#)
108. Orth, U.; Robins, R.W. Is high self-esteem beneficial? Revisiting a classic question. *Am. Psychol.* **2022**, *77*, 5–17. [\[CrossRef\]](#)

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